FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – VIII
PAPER –1
TEST DATE: 05-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –03, 18 – 20, 35 – 37):* This section contains **NINE (9)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(04 – 07, 21 – 24, 38 – 41)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(08 – 11, 25 – 28, 42 – 45):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (P), (Q), (R) and (S) and **FIVE** statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(12 – 17, 29 – 34, 46 – 51):* This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

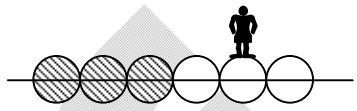
Physics

PART – I

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. The raft is a rigid structure of six logs(cylinder in shape) of the same length and diameter: three from wood of one density (shaded in the figure) having mass of each is M and three from wood of a different density (not shaded) having mass of each is m. When a person of mass m_0 stands on the second log from the edge, the raft takes a horizontal position, and each log is half immersed in water (see figure). The same one log, taken separately from the raft, holds the person standing on it, completely immersed in the water. Choose the correct alternatives



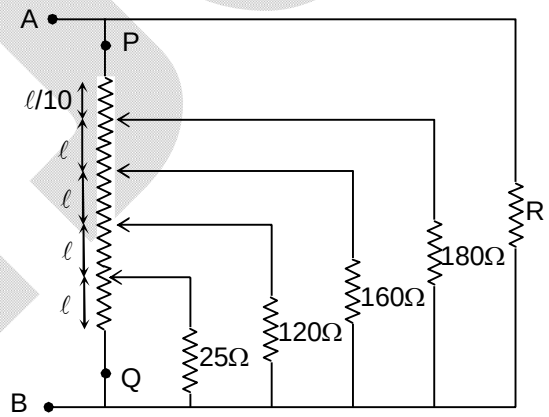
(A) The value of $\frac{M}{m}$ is 2.

(B) The value of $\frac{M}{m}=\frac{1}{2}$

(C) The value $\frac{(3M+3m)}{m_0}=3$

(D) The value $\frac{(3M+3m)}{m_0}=\frac{1}{3}$

2. The resistance of wire PQ is 410 Ω, which is tapped with the external wire as shown in the figure. The value of R is not known. When a battery of 400 V is connected across AB, the net current through the source battery is 6 ampere. Then which of the following is/are correct.



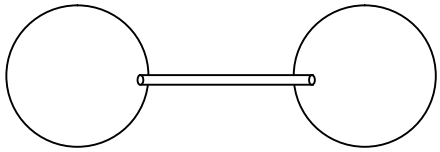
(A) The resistance of unknown resistor R is 200Ω

(B) The equivalent resistance across AB is $\frac{200}{3}\Omega$

(C) The current in 25 Ω resistor is $\frac{2}{5}$ ampere

(D) The potential drop across 120 Ω resistor is 60 volts

3. Two spherical soap bubbles in vacuum are connected through a narrow tube. Radius of left bubble is R_0 and that of other is $2R_0$. Air flows from left to right very slowly. At any instant before steady state is reached, r_1, A_1, V_1, n_1 are radius, surface area, volume and number of moles of gas in the left bubble and r_2, A_2, V_2, n_2 are same for right bubble. Assume that temperature T remains constant: Choose the CORRECT option(s). Let $n_1 = 2$ initially and S be the surface tension of soap solution.



(A) The total film area of the both the bubbles is $\frac{15RT}{S}$

(B) The rate of change of "$A_1 + A_2$" with respect to time is zero

(C) If at an instant number of moles in left bubble is 1 then $r_2 = \frac{3R_0}{\sqrt{2}}$

(D) If at an instant number of moles in left bubble is 1 then $r_1 = \frac{R_0}{\sqrt{2}}$

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

4. An airplane flies horizontally at a height H with a constant speed u. A rocket launches from the Earth from a point directly below the plane. The rocket flies at a constant speed v, and the velocity vector is always directed towards the aircraft. Neglecting the force of air resistance, find the magnitude of the thrust force of the rocket engine immediately after launch. Rocket mass m.

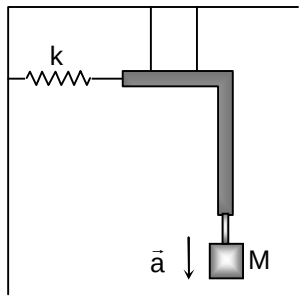
(A) $\left|\vec{F}_m\right| = \frac{1}{2}m\sqrt{g^2 + \left(\frac{vu}{H}\right)^2}$

(B) $\left|\vec{F}_m\right| = m\sqrt{g^2 + \left(\frac{vu}{H}\right)^2}$

(C) $\left|\vec{F}_m\right| = m\sqrt{g + \left(\frac{vu}{H}\right)}$

(D) $\left|\vec{F}_m\right| = m\left\{g^2 + \left(\frac{vu}{H}\right)^2\right\}$

5. The smooth tube curved at right angles is fixed so that one of its ends is directed vertically down as shown in the figure. Inside the tube there is a homogeneous flexible rope with a length L, whose diameter is slightly smaller than the diameter of the tube. The upper end of the rope through the weightless AB thread is connected to a light spring, the other end of which is fixed so that its axis is horizontal and coincides with the thread. To the lower end of the rope, without pulling it, the weight was attached to the lower end of the rope M. After releasing the load without the initial speed, it moves for a while with constant acceleration a. Find the stiffness of the spring k.



(A) $k = \frac{2mg}{L}$

(B) $k = \frac{mg}{L}$

(C) $k = \frac{mg}{2L}$

(D) $k = \frac{3mg}{L}$

6. Three identical balls with identical masses m and charges q, connected by three threads, form a regular triangle with sides ℓ. One of the threads is burned out. After some time, the balls form a right triangle. Determine the speed of the ball (modulo) located at the right angle of this triangle. There is no gravity.

(A) $v = \sqrt{\frac{q^2(\sqrt{2}-1)}{6\pi\varepsilon_0 m\ell\sqrt{2}}}$ (B) $v = \sqrt{\frac{q^2(\sqrt{2}-1)}{12\pi\varepsilon_0 m\ell\sqrt{2}}}$

(C) $v = \sqrt{\frac{q^2(\sqrt{2}-1)}{4\pi\varepsilon_0 m\ell\sqrt{2}}}$ (D) $v = \sqrt{\frac{q^2(\sqrt{2}-1)}{3\pi\varepsilon_0 m\ell\sqrt{2}}}$

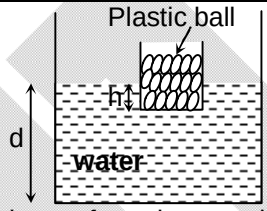
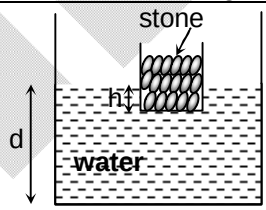
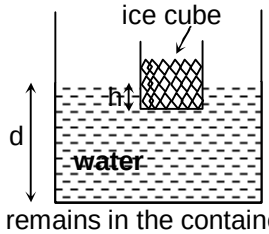
7. An organ pipe has one end closed and at the other end is a vibrating diaphragm, which is a displacement antinode. When the frequency of diaphragm is 2000 Hz, a stationary wave pattern is set up in the tube. The distance between adjacent nodes is 8.0 cm. As the frequency is slowly reduced, the stationary wave pattern disappears, but another stationary wave pattern reappears at frequency 1200 Hz calculate the length of the tube.

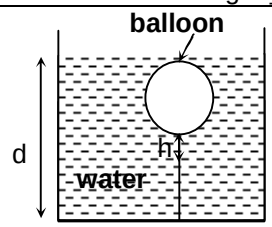
(A) 20 cm (B) 40 cm
(C) 30 cm (D) 60 cm

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

8. List–I shows different system as describe, with some parameter while list-II gives the in the parameter.

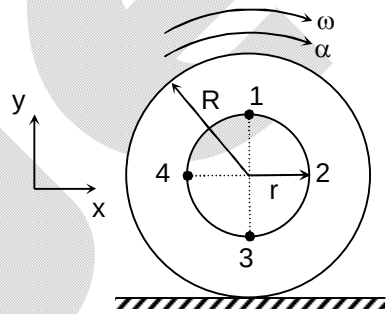
List –I		List –II	
(P)	 Plastic ball is thrown from the container in water [d is the height of water level while h is the depth to which the container is submerged]	(1)	h decreases
(Q)	 Stone is thrown from the container in the water [d is the height of water level while h is the depth to which the container is submerged]	(2)	h remain same
(R)	 Ice melts and remains in the container	(3)	h increases

	[d is the height of water level while h is the depth to which the container is submerged]		
(S)	 The water is heated slowly. [d is the height of water level while h is the tension in the string. $\gamma_{air} > \gamma_{water}$]	(4)	d decreases
		(5)	d remain same

The correct option is:

(A) (P) → (1, 5) (Q) → (1, 4) (R) → (2, 5) (S) → (3)
(B) (P) → (1, 5) (Q) → (2, 4) (R) → (2, 3) (S) → (3, 1)
(C) (P) → (1, 5) (Q) → (1, 3) (R) → (2, 5) (S) → (3)
(D) (P) → (1, 5) (Q) → (1, 5) (R) → (2, 5) (S) → (3)

9. A disc of radius R is rolling without slipping with an angular acceleration α, on a horizontal plane. Four points are marked at the end of horizontal and vertical diameter of a circle of radius r(<R) on the disc. If horizontal and vertical direction are chosen as x and y axis as shown in the figure, then acceleration of points 1, 2, 3 and 4 are $\vec{a}_1, \vec{a}_2, \vec{a}_3$ and $\vec{a}_4$ respectively, at the moment when angular velocity of the disc is ω. Match the following.

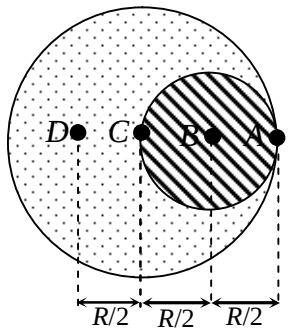


List –I		List –II	
(P)	$\vec{a}_1$	(1)	$(R\alpha - r\alpha)\hat{i} + (r\omega^2)\hat{j}$
(Q)	$\vec{a}_2$	(2)	$(R\alpha + r\alpha)\hat{i} - (r\omega^2)\hat{j}$
(R)	$\vec{a}_3$	(3)	$(R\alpha - r\alpha)\hat{i} - (r\omega^2)\hat{j}$
(S)	$\vec{a}_4$	(4)	$(R\alpha + r\alpha)\hat{i} + (r\omega^2)\hat{j}$
		(5)	None of these

The correct option is:

(A) (P) → (3) (Q) → (4) (R) → (2) (S) → (1)
(B) (P) → (3) (Q) → (2) (R) → (5) (S) → (4)
(C) (P) → (2) (Q) → (1) (R) → (4) (S) → (5)
(D) (P) → (2) (Q) → (5) (R) → (1) (S) → (5)

10. Gravitational potential on the surface of an isolated uniform solid sphere of mass *M* and radius *R* is found to be V_0. A spherical cavity having radius *R*/2 is created inside the sphere which is touching the surface of original sphere. The cavity is then filled with material having density 16 times that of original sphere. *A*, *B*, *C* and *D* are consecutive points as shown in the figure each *R*/2 apart. V_A, V_B, V_C and V_D are gravitational potentials found at points *A*, *B*, *C* and *D* respectively.

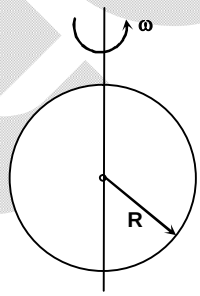


List –I		List –II	
(P)	V_A	(1)	$7V_0$
(Q)	V_B	(2)	$\frac{13}{4}V_0$
(R)	V_C	(3)	$\frac{19}{4}V_0$
(S)	V_D	(4)	$\frac{21}{4}V_0$
		(5)	$\frac{27}{4}V_0$

The correct option is:

(A) (P) → (3) (Q) → (1) (R) → (4) (S) → (2)
(B) (P) → (3) (Q) → (2) (R) → (3) (S) → (4)
(C) (P) → (4) (Q) → (1) (R) → (4) (S) → (5)
(D) (P) → (4) (Q) → (5) (R) → (1) (S) → (3)

11. Figure shows a hollow sphere (M, R) rotating with an angular speed ω. Coefficient of linear expansion is $\alpha = 10^{-5}/°C$. The temperature is increased by 10°C. List-I lists four quantities and list–II lists the corresponding values/nature of changes in these quantities.



List –I		List –II	
(P)	Kinetic energy of hollow sphere	(1)	0%
(Q)	Angular speed of hollow sphere	(2)	Increases
(R)	Angular momentum of hollow sphere	(3)	Decreases
(S)	Moment of inertia of hollow sphere	(4)	0.02%
		(5)	0.01%

The correct option is:

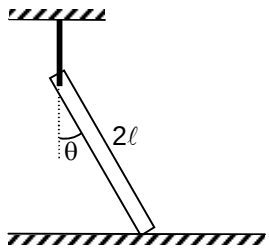
(A) (P) → (3, 1) (Q) → (3, 4) (R) → (3, 4) (S) → (2,4)
(B) (P) → (3, 4) (Q) → (3, 4) (R) → (1) (S) → (2,4)
(C) (P) → (3, 4) (Q) → (3, 5) (R) → (1) (S) → (2,1)
(D) (P) → (3, 1) (Q) → (3, 4) (R) → (2,4) (S) → (2,4)

SECTION – B

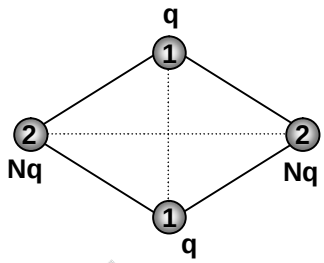
(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

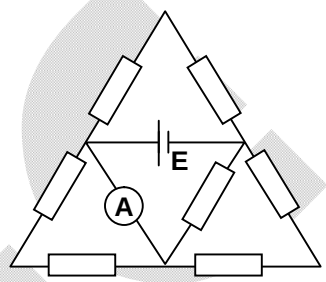
12. A uniform rod of length 2ℓ and mass m is suspended from one end by inextensible string and other end lies on smooth ground. The angle made by rod with vertical is $\theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$. If N_1 and N_2 represents the contact force from ground on rod just before and just after cutting string then find the ratio of $\frac{N_1}{N_2}$.



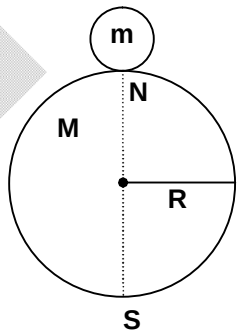
13. There were only 4 beads, which are tied on a thread at the same distance from each other so that it turned out to be a necklace. During the making of the necklace, the beads became electrified, acquiring charges of the same sign q, Nq, q, Nq. As a result, the necklace hanging in zero gravity took the shape of a rhombus. Find the ratio of the diagonals of this rhombus (larger to smaller). For calculation, take N = 8.



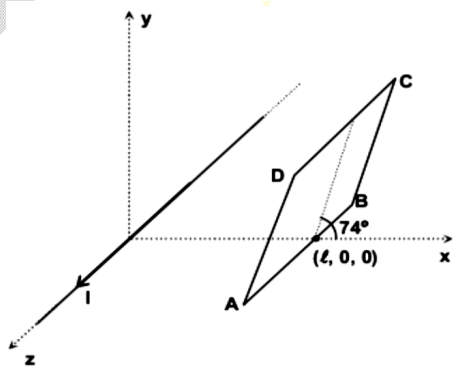
14. A triangular circuit was assembled from identical resistances R=3 Ohm each, an ideal source with emf E=2 V and an ideal ammeter (see figure). Find the ammeter reading (in ampere).



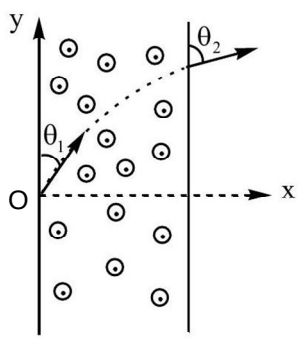
15. A satellite of mass m and radius r is projected with a velocity v_0 tangential to the surface of the earth (radius R and mass M) at the North Pole. Given r<<R. If the satellite goes to infinity, for $Rv_0^2 = xGM$. Find the value of x.



16. A wire of infinite length carrying current 'I' lies along the z-axis. A square loop of side 'ℓ' is placed such that the plane of the loop makes an angle 74° with the positive x-axis at (ℓ, 0, 0) and side AB touches the x-axis and parallel to z-axis as shown in the figure. The magnetic flux passing through the loop is $\frac{k\mu_0\ell}{\pi}$. Find the value of 100k. (Take I = 10 amp, ℓn 1.6 = 0.47 and tan 37° = 3/4)



17. Imagine non-uniform magnetic field $\vec{B} = C\left(1+\frac{x}{L}\right)(+\hat{k})$ in region of space between x = 0 to x = L as shown in figure. A particle of mass m and positive charge q enters in this region at origin. While entering and leaving it makes an angle with planes parallel to YZ as θ_1 and θ_2 respectively. Motion of charge particle is confined to xy plane. The relation $2mv(\cos\theta_2 - \cos\theta_1) = \lambda qLC$ satisfies the above said condition, find the value of λ.



Chemistry

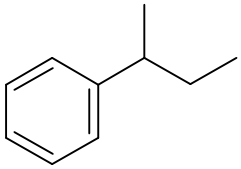
PART – II

SECTION – A
(One or More than one correct type)

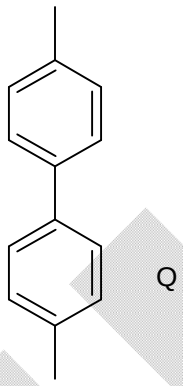
*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

18. Incorrect statements among the following is/are
(A) The most probable oxidation states for both Cr and Mo are +2, +3 and +5.
(B) Between Cr and Mo only Cr shows +6 oxidation state.
(C) Mo(VI) is more stable than Cr(VI).
(D) Both of the trioxides of Cr and Mo are amphoteric in nature.

19.



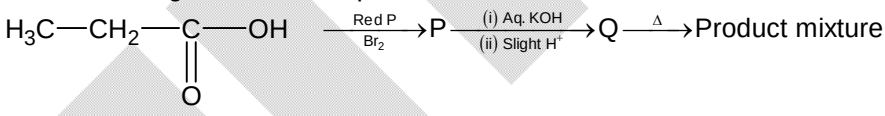
P



Q

Identify the incorrect statements about 'P' and 'Q'
(A) Both P and Q on oxidation with $KMnO_4$ produce benzoic acid.
(B) Q is inert to oxidation by $KMnO_4$
(C) Both P and Q reacts faster than benzene in electrophilic aromatic substitution reaction
(D) Q produces terepthalic acid on oxidation with $KMnO_4$

20. In the following reaction, final product mixture contains



(A) Only one optically active compound
(B) Two optically active organic compound
(C) One optically inactive compound
(D) Three optically inactive compound

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

21. What is the osmotic pressure of 0.2 M HX (aq.) solution at 300 K? $\left(\text{Given}: K_{a(HX)} = 8 \times 10^{-5}\right)$

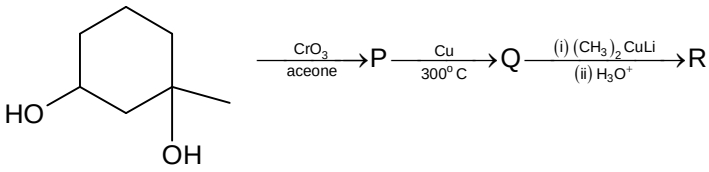
(A) 4.926 atm
(B) 4.827 atm
(C) 5.024 atm
(D) None of these

22. 0.025 mol of hydrochloride salt of a neutral amino acid is dissolved in 250 mL of water. pH of this solution is found to be 2.7. K_{a_2} of this acid is 8×10^{-8}. pH of the acid at isoelectric point would be

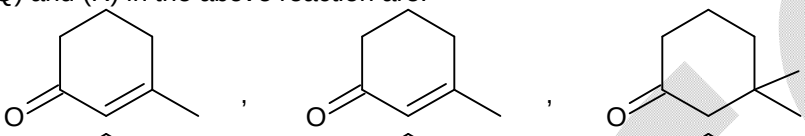
[Given log2 = 0.3]

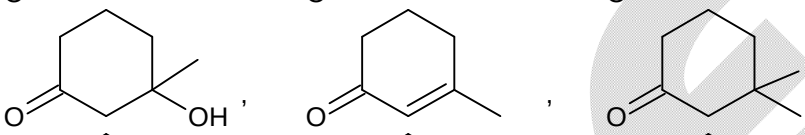
(A) 4.85 (B) 5.25

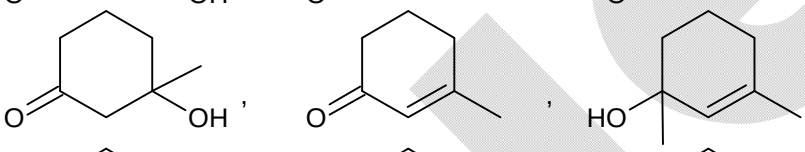
(C) 5.55 (D) 5.95

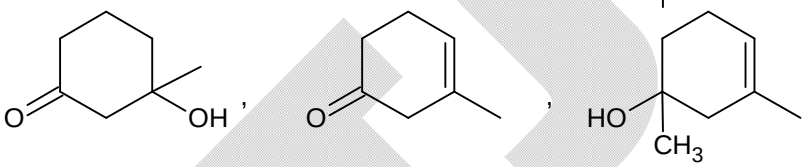
23. 

(P), (Q) and (R) in the above reaction are:

(A) 

(B) 

(C) 

(D) 

24. Which of the following ion can not be detected by the borax bead or microcosmic bead test?

(A) Cu^{2+} (B) Cr^{3+}

(C) Fe^{3+} (D) Zn^{2+}

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match the following:

List – I		List – II	
(P)	$KClO_3$	(1)	On heating producing O_2
(Q)	$KMnO_4$	(2)	Strong oxidizing agent
(R)	PbO_2	(3)	Component of safety matches
(S)	$K_2Cr_2O_7$	(4)	At least two elements in its highest oxidation state
		(5)	Component of secondary cell

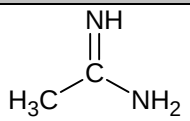
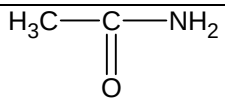
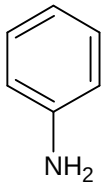
(A) P → 1, 2, 3, 4; Q → 1, 2, 5; R → 1, 2, 3, 4, 5; S → 1, 2, 3

(B) P → 1, 2, 3; Q → 1, 2, 4; R → 1, 2, 3, 5; S → 1, 2, 4

(C) P → 1, 2, 3, 4, 5; Q → 1, 2, 3; R → 1, 2, 3, 4; S → 1, 2, 4

(D) P → 1, 2, 4; Q → 1, 2, 4, 5; R → 1, 2, 3, 4, 5; S → 1, 2, 3

26. Match the following:

List – I		List – II	
(P)		(1)	Stronger base than $CH_3-CH_2-NH_2$
(Q)		(2)	Able to form salt on reacting with HCl
(R)	NH_2-NH_2	(3)	Producing a gas on boiling with NaOH
(S)		(4)	Does not give Lassaigne's test for nitrogen
		(5)	Component of rocket fuel

(A) P → 1, 2; Q → 3; R → 2, 4, 5; S → 2
(B) P → 1, 3; Q → 2, 3; R → 4, 5; S → 2
(C) P → 1, 4; Q → 2; R → 2, 3, 5; S → 4
(D) P → 1, 4; Q → 1; R → 2, 4, 5; S → 5

27. Match the following:

List – I		List – II	
(P)	White P + alkali	(1)	Non-redox reaction
(Q)	Red P + sodium chlorite	(2)	Disproportionation reaction
(R)	Phosphorous acid + PCl_3	(3)	Product contains P – P bond
(S)	Phosphoric acid + $POCl_3$	(4)	Product contains P – O – P bond
		(5)	Product is tetra protic acid

(A) P → 2; Q → 4; R → 2, 4; S → 1, 4, 5
(B) P → 2; Q → 3; R → 1, 4; S → 1, 4, 5
(C) P → 4; Q → 3; R → 1, 5; S → 1, 3, 5
(D) P → 4; Q → 3; R → 1, 5; S → 1, 2, 5

28. Match the following:

List – I		List – II	
(P)	Physisorption	(1)	Placing anhydrous $CaCl_2$ in water vapour
(Q)	Chemisorption	(2)	Multimolecular
(R)	Desorption	(3)	Low pressure is favourable condition
(S)	Absorption	(4)	$\Delta H > 0$
		(5)	Placing finally divided Fe in a closed vessel containing N_2 gas at 773 K

(A) P → 3; Q → 2; R → 3, 5; S → 1, 3
(B) P → 5; Q → 3; R → 1, 4; S → 1, 2
(C) P → 2; Q → 5; R → 3, 4; S → 1, 4
(D) P → 4; Q → 1; R → 2, 4; S → 1, 5

SECTION – B

(Numerical Answer Type)

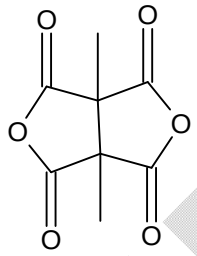
*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

29. An element E can crystallizes in fcc lattice as well as bcc lattice. If density of bcc lattice is $\frac{3\sqrt{3}}{\sqrt{2}}\text{g/cm}^3$ then the density of the fcc lattice in g/cm^3 would be_____

30. If the minimum mass of $SO_3(g)$ required to produce at least 6.4 kg of each product in the reaction given below is x kg then value of $\frac{x}{8}$ would be
(Assume % yield of reaction is 50%)
$2SO_3(g) \longrightarrow 2SO_2(g) + O_2(g)$

31. If the amount of CH_4 (in g) produced when 4.6 g of glycerol is treated with excess CH_3MgBr is x; then value of $\frac{5x}{3}$ would be

32. If number of stereoisomers in the complex $\left[M_{ABCDEF}\right]$ is x; and number of stereoisomers in the complex $\left[M_{A_2BCDE}\right]$ is y then value of $\frac{x-y}{3}$ would be

33. 

$\xrightarrow[(ii)\ \Delta]{(i)\ H_3O^+} B + C(\text{gas})$

A

The amount of 'C' gas (in g) which can make lime water milky is _____g; when we start with 0.25 mol of A. (Assume 100% yield)

34. If permissible values of m_L changes from $-2\ell,.....0,.....+2\ell$ (all integral values); then maximum number of electrons in 3d sub-shell is x; the number of Balmer lines when electron(s) de-excite from 6th excited state to ground state is y. Then value of $\frac{x \times y}{15}$ would be

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

35. Consider the circle C **:** $|z - 3 - 5i| = 2$. A point P(7 + 9i) starts moving along the line $(1+i)z+(1-i)\bar{z}+4=0$ away from the circle at a speed of $\sqrt{2}$ units/sec. A pair of tangents are drawn from P to the circle C, touching it at A and B. At any point on the arc $\overset{\frown}{AB}$ a tangent drawn to the circle intersects PA and PB at R and S. Let $\phi(t)$ denote the perimeter of ΔPRS at instant 't'. If $\phi(2) - \phi(0) = 4\left(\sqrt{a}-\sqrt{b}\right)$, then which of the following options is/are CORRECT?

(A) a and b are prime numbers
(B) a and b are coprime
(C) a – b is a composite number
(D) $\left[\frac{a-b}{5}\right]$ is a prime number

36. Let z be complex number satisfying $\text{Arg}\left(z^3-\omega^2\right)=\frac{\pi}{2}+\text{Arg}\left(z^3-\omega\right)$ where 'ω' is complex cube root of unity. Let the minimum value of $|z+2i||z+\sqrt{3}-i||z-\sqrt{3}-i|$ be $\frac{1}{2}\left(\sqrt{\alpha}-\sqrt{\beta}\right)$, then which of the following statements is/are CORRECT?

(A) α and β are prime numbers
(B) Largest prime dividing $\alpha + \beta$ is 5
(C) Largest prime dividing $\alpha + \beta$ is 13
(D) The last two digits of $3^{\alpha+\beta}$ is 01

37. Let $f(x) = \cos^{-1}(16x^5 - 20x^3 + 5x)$, then which of the following statements is true about f(x)?

(A) f(x) has 4 points of inflection
(B) f(x) is non-differentiable at 4 points
(C) $f(x) = 4\tan^{-1} x$ has 3 distinct roots in $\left[\frac{1}{2024}, \frac{2023}{2024}\right]$
(D) $f(x) = k$, $k \in [0, 3]$ has at most 5 solutions

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

38. Let $a=\sum_{r=0}^{1011}\tan^2\frac{(2r+1)\pi}{4048}$ and $b=\prod_{r=0}^{1011}\tan^2\frac{(2r+1)\pi}{4048}$, then the value of $1012\times 2023\frac{b}{a}$ is equal to

(A) 4
(B) 3
(C) 2
(D) 1

39. In a battlefield an armour piercing bullet is fired on a tank armour of thickness 20 cm. The bullet strikes the armour plate of the tank at a speed of 500 m/sec and exit the armour at 100 m/sec. Let t be the time available for the tank crew to react, then integral part of $10^4 \cdot t$ is equal to
(Given resistive force offered by the armour is proportional to square of the speed of the bullet, ln(5) = 1.609437)

(A) 0
(B) 4
(C) 9
(D) 10

40. Let C be curve which is a singular solution of the differential equation $y = x\frac{dy}{dx} + \frac{dy}{dx} - \left(\frac{dy}{dx}\right)^3$, then the area bounded by C with x = 2 is

(A) $\frac{12}{5}$ (B) $\frac{24}{5}$

(C) $\frac{36}{5}$ (D) $\frac{27}{4}$

41. A child moves one unit distance for each step he takes. He can move forward, backward, up or down either parallel to x-axis or y-axis. If he starts from (0, 0), then what is the probability that he is at (2, 2) at the end of 6 steps?

(A) $\frac{15}{256}$ (B) $\frac{15}{128}$

(C) $\frac{15}{512}$ (D) $\frac{75}{1024}$

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

42. Let $\vec{a}$ and $\vec{b}$ be two unit vectors which are orthogonal. Two lines $L_1 : \vec{r} = 6\vec{a} - \vec{b} + \lambda(2\vec{b} - \vec{a})$ and $\vec{r} = \vec{a} - \vec{b} + \mu(\vec{a} + 3\vec{b})$ where $\lambda, \mu \in R$ are coplanar, then choose the correct option from the following List(s) (where [.] represents greatest integer function)

List – I		List - II	
(P)	$\lambda + \mu$ is equal to	(1)	1
(Q)	$\lvert\lambda - \mu\rvert$ is equal to	(2)	5
(R)	If θ is angle between L_1 and L_2, then [tan θ] + 3	(3)	6
(S)	Distance of point of intersection of L_1 and L_2 from origin is 'd', then [d] is	(4)	4
		(5)	0

The correct option is:

(A) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (2)

(B) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (2)

(C) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (1); (S) $\rightarrow$ (5)

(D) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (4); (S) $\rightarrow$ (5)

43. Match the following List-I with List-II

List – I		List - II	
(P)	If $f_1(n)$ be the number of divisors of n and $f_k(n) = f_1(f_{k-1}(n))$, then the least integer k for which $f_k(2024^{2024}) = 3$ is	(1)	2
(Q)	Let A be a non-singular matrix such that $A^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 6 & 8 \\ 100 & 100 & 99 \end{bmatrix}$, then the value of $\frac{1}{3}\{\det(\text{adj}(2A)) + \det(2A)\}$	(2)	1
(R)	Tangents AB and AC are drawn from A to $x^2 + y^2 - 2x + 4y + 1 = 0$ meeting it at B and C. If equation of circumcircle of ΔABC is $x^2 + y^2 - x + y - 2 = 0$. Such that A is (a, b), then a + b is	(3)	4
(S)	The value of $\left[\tan\frac{\pi}{7}\tan\frac{2\pi}{7}\tan\frac{3\pi}{7}\right]$ is (where [.] represents greatest integer function)	(4)	6
		(5)	0

The correct option is:
(A) (P) → (1); (Q) → (3); (R) → (2); (S) → (5)
(B) (P) → (4); (Q) → (5); (R) → (3); (S) → (1)
(C) (P) → (4); (Q) → (3); (R) → (2); (S) → (1)
(D) (P) → (1); (Q) → (3); (R) → (4); (S) → (5)

44. Match the following List-I with List-II

List – I		List – II	
(P)	If α, β, γ are roots of the equation $x^3 - x^2 - 2x - 3 = 0$ and $\Delta = \begin{vmatrix} \beta^2 & 2\beta\gamma - \alpha^2 & \gamma^2 \\ \alpha^2 & \gamma^2 & 2\alpha\gamma - \beta^2 \\ 2\alpha\beta - \gamma^2 & \beta^2 & \alpha^2 \end{vmatrix}$, then $\sqrt{\Delta}$ is	(1)	1
(Q)	Let $f(k) = \int_0^4 \lvert x(4-x) - k \rvert dx$ be minimum, then the value of k is	(2)	3
(R)	A chord AB of an ellipse $\frac{x^2}{9} + \frac{y^2}{\lambda^2} = 1$ subtends 90° at centre of this ellipse 'O'. If $\frac{1}{OA^2} + \frac{1}{OB^2} = \frac{14}{45}$, then λ^2 is equal to	(3)	5
(S)	Let $L = \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{\sqrt[2025]{n^{2025} + r \cdot 2025}}$, then the value of L is	(4)	7
		(5)	0

The correct option is:
(A) (P) → (4); (Q) → (3); (R) → (5); (S) → (2)
(B) (P) → (5); (Q) → (3); (R) → (2); (S) → (1)
(C) (P) → (5); (Q) → (3); (R) → (1); (S) → (2)
(D) (P) → (4); (Q) → (3); (R) → (2); (S) → (1)

45. If g(x) is inverse of f(x), then match the following List-I with List-II

List – I		List – II	
(P)	$f(x) = 2x + \cos x$, then $\frac{1}{4g''(1)}$ is	(1)	–1
(Q)	$f(x) = x^x$, then $\frac{1}{2}g''(1)$ is	(2)	0
(R)	$f(x) = \tan^{-1} x$, then $g''(0)$ is	(3)	1
(S)	$f(x) = e^{x^3+x}$, then $\lvert g''(1)\rvert$	(4)	2
		(5)	8

The correct option is:
(A) (P) → (4); (Q) → (1); (R) → (2); (S) → (3)
(B) (P) → (1); (Q) → (5); (R) → (2); (S) → (3)
(C) (P) → (1); (Q) → (3); (R) → (4); (S) → (5)
(D) (P) → (4); (Q) → (1); (R) → (2); (S) → (5)

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

46. If the sum of the series $\frac{3(3-2^2)}{3^2-2\cdot3+2^2}+\frac{3(2^2\cdot3^2-2^5)}{3^4-2^2\cdot3^2+2^4}+\frac{3(2^5\cdot3^4-2^{10})}{3^8-3^4\cdot2^4+2^8}+.....$ upto infinite terms is $\frac{a\cdot b}{c}$ where a, b, c are prime numbers, then the value of a·b – c is equal to

47. Let $y = (x^2 + 3x)\sin x$. If $\frac{d^{10}y}{dx^{10}} = P(x)\cdot\sin x + Q(x)\cos x$, then the value of $\lvert P'(1)\rvert$ is equal to

48. Let $I_1 = \int_0^{\frac{\pi}{2}} \frac{(\log_e \tan\theta)^2}{1-\sin 2\theta}d\theta$ and $I_2 = \int_0^{\frac{\pi}{2}} \log(1-\sin\theta)\cot\theta\, d\theta$, then $\left|\frac{I_1}{I_2}\right| = ?$

49. Let A, B, C be acute angles such that $\begin{vmatrix} \cos^3 A & (\cos A+2)^3 & (\cos A-2)^3 \\ \cos^3 B & (\cos B+2)^3 & (\cos B-2)^3 \\ \cos^3 C & (\cos C+2)^3 & (\cos C-2)^3 \end{vmatrix} = 0$, then the value of $8\cdot\sum \sec A \sec B$ is equal to

50. Let f(x) be non-negative continuous function on R such that $f(x)+f\left(x+\frac{1}{2}\right)=3$. Let $n = \int_0^{2024} f(x)dx$ and m be the number of natural numbers ≤ n which are coprime with n, then the value of $\left[\frac{m}{2024}\right]$ (where [.] represents greatest integer function)

51. Consider the equation $x^4 + 2ax^3 + (2a-2)x^2 + (3-4a)x - 2 = 0$. A die is thrown to select values of 'a'. If probability that this equation has exactly 2 real roots is P, then the largest prime divisor of $\frac{1}{P}$ is equal to